

VIDEO GAME GLOBAL SALES PREDICTION

Final Project Report

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1. Title

Video Game Global Sales Prediction Using Machine Learning

2. Abstract

The video game industry generates large amounts of sales data across different platforms, genres, publishers, and release years. Predicting global video game sales can support data-driven analysis of commercial performance. This project presents an end-to-end machine learning approach using the Video Game Sales dataset obtained from Kaggle. Data preprocessing was performed to handle missing values and duplicate records, followed by exploratory data analysis and feature selection. Platform, Year, Genre, and Publisher were used as predictive features, while Global_Sales was the target variable. Four regression models—Linear Regression, Ridge Regression, Random Forest Regression, and Gradient Boosting Regression—were developed and evaluated using MSE, MAE, RMSE, and R^2 . Gradient Boosting Regression achieved the best performance among the tested models, with an R^2 of approximately 0.1015. The selected model was saved and used for prediction on new and unseen game data. A Streamlit application was also prepared to demonstrate practical deployment.

3. Keywords

Video Game Sales; Machine Learning; Regression; Global Sales Prediction; Gradient Boosting; Random Forest; Linear Regression; Exploratory Data Analysis; Predictive Analytics

4. Introduction

The video game industry has grown significantly over the past several decades, resulting in large volumes of data related to games, platforms, genres, publishers, and sales. Machine learning provides techniques for identifying patterns in historical data and using them to make predictions for new observations. Regression algorithms are particularly useful when the target variable is continuous, such as global video game sales. This project applies machine learning techniques to predict Global_Sales using selected attributes from the Video Game Sales dataset and follows an end-to-end workflow from data understanding to final prediction and deployment.

5. Problem Statement

The objective is to develop a machine learning model capable of predicting the Global_Sales of a video game based on Platform, Year, Genre, and Publisher. Since Global_Sales is a continuous numerical variable, the task is formulated as a supervised regression problem.

6. Research Objectives

- Understand the dataset.
- Preprocess and clean the dataset.
- Identify useful predictive features.
- Perform exploratory data analysis.
- Develop multiple regression models.
- Evaluate and compare the models.

- Select the best-performing model.
- Perform prediction on new and unseen data.
- Develop a reusable prediction application.

7. Research Questions

- Can machine learning predict global video game sales?
- Which available features are useful?
- Which regression algorithm performs best?
- How do the four selected models compare?
- Can the selected model predict unseen game information?
- What are the limitations of the available data?

8. Literature Review / Related Work

Previous studies in video game analytics have investigated relationships between game characteristics and commercial success. Machine learning methods are increasingly used for sales prediction, recommendation, player behavior analysis, and market analysis. Linear Regression provides a simple and interpretable baseline, Ridge Regression adds L2 regularization, and tree-based ensemble methods such as Random Forest and Gradient Boosting can model nonlinear relationships. This project compares these approaches on a common dataset.

9. Research Gap

Video game sales are influenced by many factors. The available dataset does not contain variables such as marketing expenditure, game ratings, user reviews, developer reputation, social media popularity, or pre-release interest. Therefore, a gap exists between the structured historical attributes available in the dataset and the broader factors influencing commercial success. This study evaluates the predictive information available from selected features.

10. Dataset Description

The project uses the Video Game Sales dataset from Kaggle. It contains historical information about video games released across different platforms, including game identity, platform, year, genre, publisher, regional sales, and global sales. Selected inputs are Platform, Year, Genre, and Publisher and the target is Global_Sales.

Dataset Feature Table

Feature	Description
Rank	Ranking based on sales
Name	Video game title
Platform	Platform on which the game was released
Year	Release year
Genre	Game genre
Publisher	Game publisher
NA_Sales	Sales in North America
EU_Sales	Sales in Europe

JP_Sales	Sales in Japan
Other_Sales	Sales in other regions
Global_Sales	Worldwide sales in millions of units

11. Methodology

The methodology consists of dataset collection and understanding, preprocessing, exploratory analysis, feature selection, train-test splitting, categorical encoding, numerical scaling, model development, evaluation, comparison, final model selection, and unseen-data prediction.

12. System / ML Workflow

Dataset Collection → Dataset Understanding → Data Preprocessing → EDA → Feature Selection → Train-Test Split → Data Transformation → Model Development → Model Evaluation → Model Comparison → Final Model Selection → Unseen Prediction → Deployment

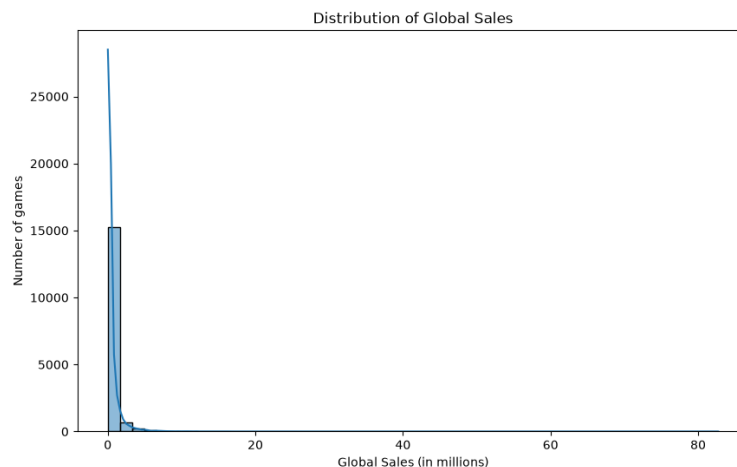
13. Data Preprocessing

Missing values in relevant columns were handled using appropriate methods. Missing Year values were replaced using the median, while missing Publisher and Genre values were filled using their modes. Duplicate records were removed. Extreme sales values were not blindly removed because high-selling games represent legitimate observations. Categorical variables were transformed using One-Hot Encoding and Year was standardized using StandardScaler.

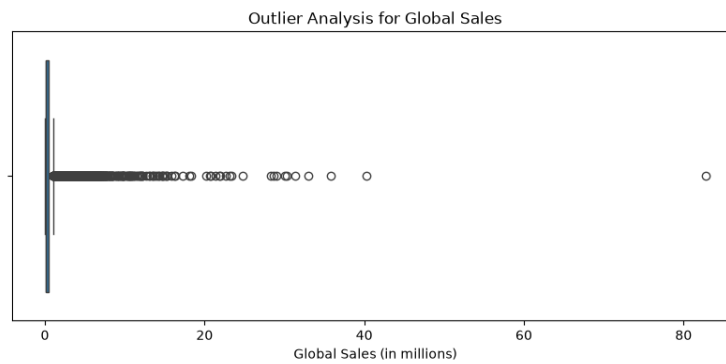
14. Exploratory Data Analysis (EDA)

EDA was performed to understand global sales distribution, sales across platforms and genres, sales trends over years, and publisher-related patterns. Suitable graphs and statistical summaries were used to understand the dataset before model development.

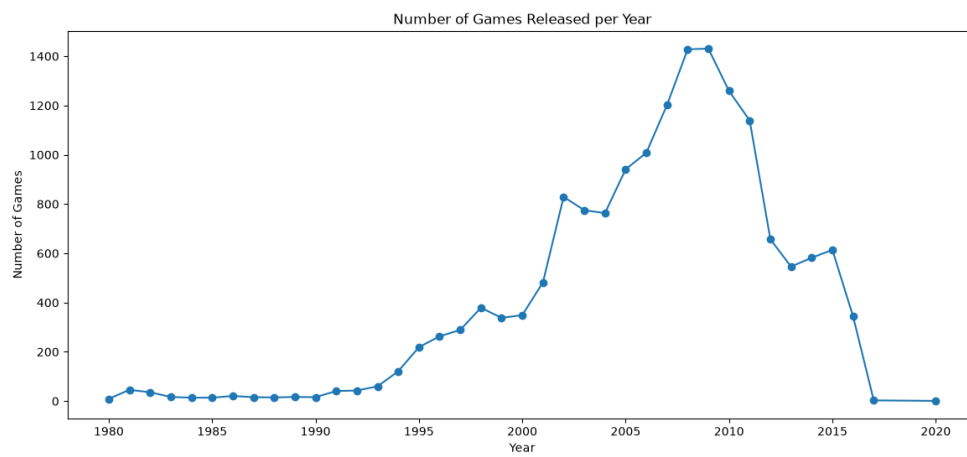
Distribution of Global Sales



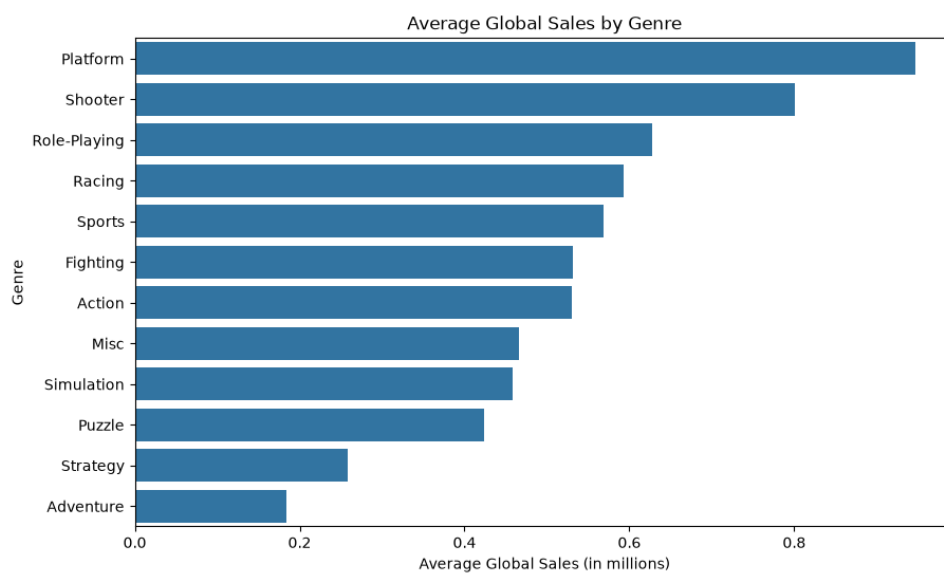
Outlier Analysis for Global Sales



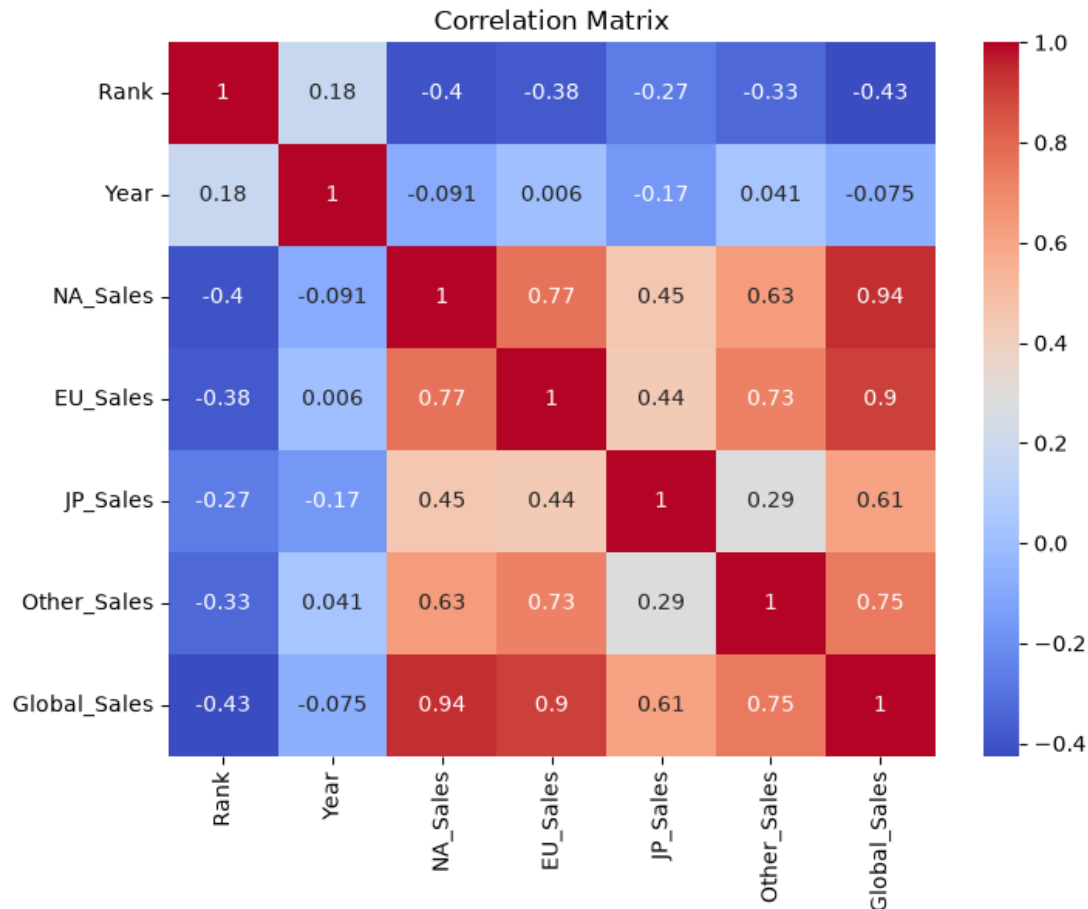
Number of Games Released per Year



Average Global Sales by Genre



Correlation Matrix



15. Feature Engineering / Feature Selection

Platform, Year, Genre, and Publisher were selected as input features. Global_Sales was separated as the target. One-Hot Encoding was applied to categorical features and StandardScaler was applied to Year.

16. Model Development

Four regression models were developed: Linear Regression, Ridge Regression, Random Forest Regression, and Gradient Boosting Regression. These models provide baseline, regularized linear, bagging ensemble, and boosting approaches for comparison.

Model Summary

Model	Purpose
Linear Regression	Baseline linear model
Ridge Regression	L2-regularized linear model
Random Forest Regression	Tree-based ensemble model
Gradient Boosting Regression	Sequential boosting model

17. Experimental Setup

The implementation was performed in Python using Visual Studio Code and Jupyter Notebook. Pandas and NumPy were used for data handling, Matplotlib and Seaborn for visualization, Scikit-learn for machine learning, Joblib for model persistence, and Streamlit for deployment. An 80:20 train-test split with `random_state=42` was used.

18. Model Validation

The dataset was divided into 80% training data and 20% testing data. The training set was used to learn model parameters, while the test set remained separate for evaluating performance on unseen observations. The same split was used for model comparison.

19. Hyperparameter Tuning

Hyperparameter tuning is the process of selecting suitable values for the parameters that control the behavior of a machine learning algorithm. In this project, hyperparameter tuning was considered for the regression models. However, extensive tuning was not applied, because hyperparameter tuning should be used only when required. The primary objective of this project was to compare multiple suitable regression models and select the best-performing model based on evaluation metrics.

The models were trained using reasonable default or manually selected parameter values. The performance of each model was compared using MSE, MAE, RMSE, and R^2 Score. Based on this comparison, Gradient Boosting Regression was selected as the final model because it achieved the best overall performance.

20. Model Evaluation

MSE, MAE, RMSE, and R^2 Score were used. MSE measures squared prediction error, MAE measures average absolute error, RMSE expresses error in the target unit, and R^2 measures the proportion of target variance explained by the model.

21. Model Comparison

Model	MSE	MAE	RMSE	R^2
Linear Regression	3.8926	0.5662	1.9730	0.0889
Ridge Regression	3.8908	0.5637	1.9725	0.0893
Random Forest Regression	4.1929	0.5358	2.0477	0.0186
Gradient Boosting Regression	3.8388	0.5244	1.9593	0.1015

22. Final Model Selection

Gradient Boosting Regression was selected as the final model because it produced the strongest overall metrics among the tested models. The trained model was serialized using Joblib and stored as `final_model.pkl` for reuse without retraining.

23. Sample / Unseen Data Prediction

The saved model was used to predict the global sales of a new game with Platform = PS4, Year = 2015, Genre = Action, and Publisher = Sony Computer Entertainment. Insert the exact prediction generated by the final `predict.py` run in the table below.

Sample Prediction Table

Feature	Sample Value
Platform	PS4
Year	2015
Genre	Action
Publisher	Sony Computer Entertainment
Predicted Global Sales	1.20 million units

24. Results

Gradient Boosting Regression performed best among the four tested models. Its reported metrics were $MSE = 3.8388$, $MAE = 0.5244$, $RMSE = 1.9593$, and $R^2 = 0.1015$. The trained model was saved and successfully used for new-data prediction. A Streamlit application was prepared to provide a user-friendly prediction interface.

25. Discussion

The results demonstrate that machine learning can be applied to video game sales prediction, but the relatively low R^2 indicates that the selected features explain only a limited part of the variation in global sales. Similar performance of Linear and Ridge Regression suggests limited linear predictive information, while Gradient Boosting performed comparatively better, indicating possible nonlinear relationships.

26. Limitations

- The dataset contains historical sales information.
- Marketing expenditure is unavailable.
- Game ratings and reviews are not included.
- Developer reputation and game budget are not considered.
- Social media popularity and search trends are unavailable.
- The relatively low R^2 indicates limited predictive power.

27. Conclusion

This project successfully implemented an end-to-end machine learning pipeline for predicting global video game sales. The workflow covered data understanding, preprocessing, EDA, feature selection, model

development, evaluation, comparison, final model selection, and unseen-data prediction. Four regression algorithms were evaluated, and Gradient Boosting Regression achieved the best performance with an R^2 of approximately 0.1015. The model was saved as a reusable artifact and integrated into a simple Streamlit deployment. Although performance is limited by the available features, the project demonstrates a complete and explainable machine learning development process.

28. Future Scope

- Include game ratings and user reviews.
- Include marketing and advertising expenditure.
- Include developer information, game budget, and franchise history.
- Include release month and seasonal effects.
- Use social media engagement and search popularity.
- Experiment with additional ensemble and advanced regression algorithms.
- Perform systematic hyperparameter optimization when justified.
- Use larger and more recent datasets.

29. References

1. Kaggle. Video Game Sales Dataset, Gregor Ustinov.
2. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
3. James, G., Witten, D., Hastie, T., & Tibshirani, R. *An Introduction to Statistical Learning*. Springer.
4. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
5. Pandas documentation – Data analysis and manipulation in Python.
6. Matplotlib and Seaborn documentation – Data visualization.
7. Streamlit documentation – Python application deployment.

30. Appendix

A. Project Structure

```
videogamesales_prediction/  
├── data/raw/vgsales.csv  
├── data/processed/vgsales_cleaned.csv  
├── models/final_model.pkl  
├── notebooks/vgsales.ipynb  
├── outputs/model_comparison.csv  
├── outputs/new_game_prediction.csv  
├── src/preprocess.py  
├── src/train.py  
├── src/evaluate.py  
├── src/predict.py  
├── tests/test_project.py  
├── app.py  
└── requirements.txt
```

B. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib, Jupyter Notebook, Streamlit, and VS Code.

C. Final Model

Gradient Boosting Regression — $MSE \approx 3.8388$; $MAE \approx 0.5244$; $RMSE \approx 1.9593$; $R^2 \approx 0.1015$.

D. Sample Prediction

Platform: PS4 , Year: 2015 , Genre: Action , Publisher: Sony Computer Entertainment.

 Predict Global Sales

Prediction completed successfully!



Prediction Result

Estimated Global Sales: 1.20 million units



Game Details

	Platform	Year	Genre	Publisher
0	PS4	2015	Action	Sony Computer Entertainment